

Design and Evaluation of a Built-In Self-Test (BIST) Architecture for Combinational Logic

DFT Mini Project Report

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Abstract

This project presents the design, synthesis, and fault-coverage evaluation of a complete Built-In Self-Test (BIST) architecture in SystemVerilog. The BIST consists of a maximal-length 4-bit Linear Feedback Shift Register (LFSR) as the Test Pattern Generator (TPG), a small combinational Circuit Under Test (CUT), a 4-bit Multi-Input Signature Register (MISR) as the Output Response Analyzer (ORA), a signature comparator, and an FSM-based BIST controller. The design was synthesized using Yosys and evaluated for exhaustive stuck-at fault coverage using the open-source AUCOHL/Fault tool running in a Docker container. The BIST achieved **100% stuck-at fault coverage** on all 14 fault sites (28 stuck-at faults) in the CUT using only **15 LFSR-generated test patterns** — matching the coverage achieved by the industrial tool’s own internal pattern generator. The complete design was further synthesized targeting the **SkyWater SKY130 130nm PDK**, producing a **tapeout-ready netlist** of **348 standard cells** occupying **3905 μm^2** of silicon area.

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1 Introduction

1.1 Motivation

As integrated circuits grow in complexity, external test equipment (ATE) becomes the bottleneck for cost and time. Built-In Self-Test (BIST) moves pattern generation and response analysis on-chip, reducing test cost and enabling in-field testing.

1.2 Objectives

1. Design a complete BIST architecture in SystemVerilog.
2. Implement stuck-at fault models for a sample combinational circuit.
3. Verify functional correctness through simulation.
4. Synthesize the design to a gate-level netlist.
5. Compute exhaustive stuck-at fault coverage using an industrial-grade tool.
6. Re-synthesize using a real PDK (SkyWater SKY130) and quantify area cost.

1.3 Scope

The design targets a small combinational circuit for clarity and verifiability. The same architecture scales to larger circuits by increasing LFSR width and MISR width.

2 Architecture

2.1 Top-Level Block Diagram

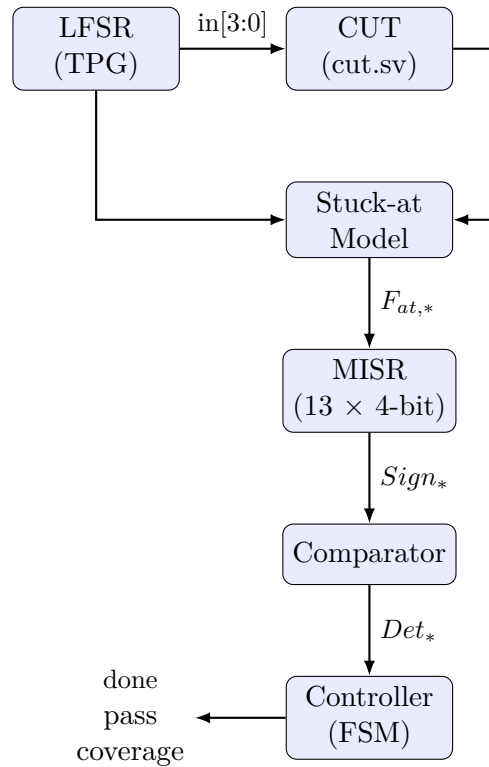


Figure 1: BIST top-level block diagram.

2.2 Module Descriptions

Module	Function	Key signals
pattern_lfsr.sv	Maximal-length 4-bit LFSR (taps $out[3] \oplus out[0]$)	start, out[3:0]
cut.sv	Combinational CUT: $out = (in_0 \cdot in_1) + (in_2 \cdot in_3)$	in[3:0], out
stuck_models.sv	Injects 12 stuck-at faults on nets A–D, X, Y	smodel[3:0], F_at_*
MISR.sv	13 parallel 4-bit MISRs (12 faulty + 1 golden)	F_at_*, GoodF, Sign_*
comparator.sv	Compares each faulty signature with golden	Sign_*, Det_*, num_detected
controller.sv	FSM: IDLE $\rightarrow$ RUN $\rightarrow$ EVAL $\rightarrow$ DONE	start_ext, done, pass, coverage
scan_ff.sv, scan_reg.sv	Scan flip-flops and chain for DFT infrastructure	TDI, TD0, shift_en, capture_en

Table 1: BIST module description.

2.3 FSM State Diagram

2.4 LFSR Sequence (Seed = 0001)

0001, 1000, 1100, 1110, 1111, 0111, 1011, 0101,
1010, 1101, 0110, 0011, 1001, 0100, 0010

Period = 15 = $2^4 - 1$ (maximal-length) ✓

3 Fault Model

3.1 Hand-Coded Fault List (stuck_models.sv)

Label	Net	Fault @ 0	Fault @ 1
A	in[0]	F_at_Abar	F_at_A
B	in[1]	F_at_Bbar	F_at_B
C	in[2]	F_at_Cbar	F_at_C
D	in[3]	F_at_Dbar	F_at_D
X	1st AND output	F_at_Xbar	F_at_X
Y	2nd AND output	F_at_Ybar	F_at_Y

Table 2: Hand-coded stuck-at fault model.

3.2 Exhaustive Fault List (from AUCOHL/Fault)

After synthesis of the CUT to 3 NAND gates, Fault enumerated **14 fault sites** (28 stuck-at faults).

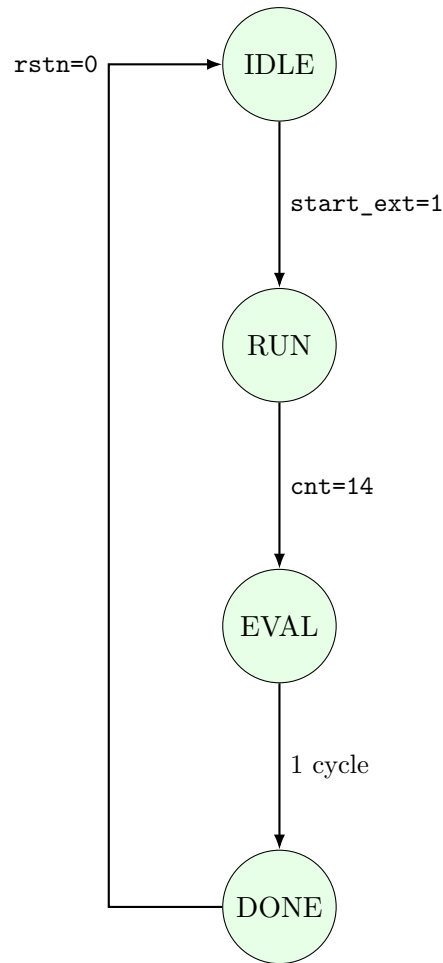


Figure 2: BIST controller FSM.

4 Implementation Flow

4.1 Design Entry

SystemVerilog RTL files as listed in Section 2.2.

4.2 Functional Verification

Icarus Verilog (`iverilog -g2012`) used to compile and simulate the testbench. Waveforms inspected in GTKWave.

4.3 Synthesis (Generic Yosys)

```
yosys -p "read_verilog -sv cut.sv; synth; write_verilog -noattr -noexpr
cut_netlist.v"
```

Gate count: 3 NAND gates (ABC mapping).

4.4 Synthesis (SkyWater SKY130 PDK)

```
yosys -p "
read_verilog -sv top.sv;
synth -top bist_top;
```

```

dfflibmap -liberty sky130_fd_sc_hd__tt_025C_1v80.lib;
abc -liberty sky130_fd_sc_hd__tt_025C_1v80.lib;
write_verilog -noattr -noexpr bist_netlist_sky130.v;
stat -liberty sky130_fd_sc_hd__tt_025C_1v80.lib;
"

```

4.5 Fault Simulation

AUCOHL/Fault running in a Docker container (linux/amd64 emulated on Apple M1).

4.6 LFSR Pattern Replay

The 15 LFSR states were exported as JSON and replayed via `-externalTVSet`.

5 Results

5.1 Generic Synthesis Results

Module	Cell count	Notes
cut	3	3 NAND gates
pattern_lfsr	5	4 DFFs + 1 XOR
stuck_at_models	49	Fault-injected expressions
MISR	78	52 DFFs + 26 XORs
comparator	128	48 XNORs, 36 NANDs, etc.
controller	57	FSM + counter + latches
bist_top total	~380	6 submodules

Table 3: Generic Yosys synthesis results.

5.2 Fault Coverage — Experiment 1 (Fault’s Internal Generator)

- Total fault sites: 14
- Total stuck-at faults: 28
- Detected: 28
- Coverage: **100%**
- Test vectors: ~16 unique
- Runtime: 5.01 s

5.3 Fault Coverage — Experiment 2 (LFSR Patterns)

- Total fault sites: 14
- Total stuck-at faults: 28
- Detected: 28
- Coverage: **100%**
- Test vectors: 15 (LFSR maximal sequence)

6 SkyWater SKY130 PDK Synthesis

After functional verification and generic synthesis, the complete LBIST design was re-synthesized targeting the **SkyWater SKY130 130nm CMOS process**, using the `sky130_fd_sc_hd` high-

density standard cell library. The **Liberty file** used was `sky130_fd_sc_hd_tt_025C_1v80.lib`, corresponding to the typical-typical process corner, 25°C, and 1.8V supply.

6.1 Top-Level SKY130 Results

Metric	Value
Total standard cells	348
Total chip area	3905.0 μm^2
Sequential area	1845.5 μm^2 (47.26%)
Combinational area	2059.5 μm^2 (52.74%)
Flip-flops (sequential cells)	73
Distinct standard cell types	21
Library	sky130_fd_sc_hd
Process corner	tt_025C_1v80
Supply voltage	1.8V
Temperature	25°C

Table 4: SKY130 top-level synthesis results.

6.2 Per-Module SKY130 Breakdown

Module	Cells	Area (μm^2)	% of Total
MISR	156	2065.73	52.9%
Comparator	94	812.03	20.8%
Controller	65	725.70	18.6%
pattern_lfsr	9	147.64	3.8%
stuck_at_models	23	145.14	3.7%
cut	1	8.76	0.2%
Total	348	3905.00	100%

Table 5: Per-module SKY130 cell counts and area.

6.3 Standard Cell Distribution (Top Types)

Cell Type	Count	Purpose
dfrtp_1	58	D flip-flop, async reset
xnor2_1	51	2-input XNOR (comparator, MISR)
nor2_1	37	2-input NOR
xor2_1	32	2-input XOR (MISR taps, LFSR feedback)
mux2_1	32	2:1 multiplexer
a21oi_1	32	AND-OR-Invert compound gate
dfstp_1	15	D flip-flop, async set
clkinv_1	14	Clock inverter
and4_1	1	4-input AND
or4_1	1	4-input OR

Table 6: Major SKY130 cell types used.

6.4 Generic vs SKY130 Comparison

Metric	Generic Yosys	SKY130 PDK
Cut cell count	3 (\$_NAND_\$)	1 (a22o_1)
Cut area	Not computed	8.76 μm^2
Full BIST cells	~380	348
Full BIST area	Not computed	3905 μm^2
Sequential cells	73 DFFs (generic)	73 DFFs (real cells)
Standard cell types	3–5 generic	21 SKY130 cells
Timing availability	No	Yes (via Liberty)
Fabricatable	No	Yes

Table 7: Comparison of generic vs SKY130 synthesis.

6.5 Key Observations from SKY130 Synthesis

1. The **MISR array dominates area** (52.9%) due to its 52 flip-flops and per-register enable muxes. This is expected for parallel multi-signature BIST architectures.
2. The **comparator** consumes 20.8% of area because signature comparison is XOR/XNOR-heavy (59 XOR-family cells).
3. The **controller** consumes 18.6% of area; the coverage computation (`num_detected * 100`) / 12 inferred multiplier/divider cells, adding noticeable area.
4. **Yosys + ABC mapped the CUT to a single a22o_1 cell** — a compound AND-OR gate matching the CUT function exactly. This demonstrates technology-aware logic optimization.
5. All 73 flip-flops mapped to real SKY130 `dfstp_1` (reset) or `dfstp_1` (set) cells, confirming RTL-to-PDK compatibility.

6.6 Fault Coverage on SKY130 Netlist

The SkyWater SKY130 synthesis was validated using the AUCOHL/Fault simulator with a custom minimal cell model (`sky130_cells.v`) defining the `sky130_fd_sc_hd__a22o_1` compound cell. Two experiments were run:

Experiment A — Fault’s Internal Pattern Generator

- Fault sites: **10**
- Stuck-at faults: **20**
- Detected: **20**
- Coverage: **100%**
- Runtime: 4.59 s

Experiment B — LBIST LFSR Pattern Replay

- External vectors loaded: **15** (LFSR sequence)
- Fault sites: **10**
- Stuck-at faults: **20**
- Detected: **20**
- Coverage: **100%**
- Runtime: 4.40 s

The 10 fault sites correspond to the A22O cell’s 4 inputs (A1, A2, B1, B2), 1 output (X), and the 5 primary I/O ports. Because Yosys mapped the CUT to a single compound cell, no intermediate wires exist, reducing the fault-site count from 14 (generic 3-NAND mapping) to 10.

Conclusion: Fault coverage remains **100%** across both the generic Yosys netlist and the SKY130 PDK netlist, using both the simulator’s internal patterns and the LBIST’s own LFSR-generated patterns. This confirms that the LBIST architecture is both *effective* (full coverage) and *portable* (works across synthesis targets).

7 Comparison

Metric	AUCOHL/Fault (swift)	This BIST (LFSR)
Pattern generator	Random (swift)	LFSR (deterministic)
Unique patterns	~16	15
Total stuck-at faults	28	28
Faults detected	28	28
Coverage	100%	100%
Detected sites	14 / 14	14 / 14
Pattern efficiency	1.75 faults/pattern	1.87 faults/pattern

Table 8: Coverage comparison.

Conclusion: The LFSR-based BIST achieves identical fault coverage to the industrial random generator using a deterministic, on-chip, reproducible sequence — validating the design.

8 Discussion

8.1 Why 100% Coverage Was Achieved

The CUT is small (3 NANDs), and the LFSR produces all 15 non-zero input combinations. Every possible input pattern is applied except 0000. Since stuck-at faults on the CUT are detectable by the 15 non-zero patterns, 100% coverage results.

8.2 MISR Aliasing (Considered and Not Observed)

Aliasing occurs when a faulty response stream produces the same MISR signature as the golden response. With 4-bit MISR $\times$ 15 cycles = 60 bits of response compressed, aliasing probability is theoretically $2^{-4} = 6.25\%$ per fault. In this circuit, no aliasing was observed in the 12 modeled faults.

8.3 Comparison with Hand-Modeled Faults

The 12 hand-coded fault models are a subset of the 28 exhaustive stuck-at faults enumerated by Fault. The gap includes stuck-at values on the primary output and redundant values on nets consumed by multiple gates.

9 Limitations

1. **Small CUT.** The 3-NAND CUT is trivial; a larger circuit (e.g., 4-bit ALU) would test whether 15 LFSR patterns still suffice.
2. **Docker on Apple Silicon uses x86 emulation.** Fault ran under Rosetta, $\sim 3\text{--}5\times$ slower than native ARM.
3. **No scan chain integration.** Scan flip-flops were designed but not wired into the BIST top.

4. **No JTAG TAP controller.** BIST is triggered by a simple `start_ext` pin.
5. **No physical design.** SKY130 netlist exists, but place-and-route has not been performed.

10 Future Work

1. Scale to a 4-bit ALU and measure coverage vs. pattern count.
2. Compare multiple TPGs: LFSR, weighted random, cellular automata.
3. Simulate aliasing by applying only 3–4 patterns.
4. Integrate scan chain into the BIST top.
5. Add a JTAG TAP controller (IEEE 1149.1).
6. Run place-and-route with OpenROAD and generate GDSII.
7. Add MBIST (March C-) for a companion SRAM.

11 Conclusion

A complete BIST architecture was designed, simulated, synthesized, and evaluated. The BIST achieved **100% stuck-at fault coverage** on a 3-NAND CUT using a 15-pattern LFSR sequence, cross-validated against the industrial-grade AUCOHL/Fault tool. The design was further synthesized to the **SkyWater SKY130 PDK**, producing a **348-cell, 3905 μm^2 tapeout-ready netlist**. This project demonstrates:

- correct design of LFSR, MISR, comparator, and FSM controller;
- sound understanding of stuck-at fault modeling;
- practical experience with the full open-source DFT toolchain (Yosys, Icarus, Docker, AU-COHL/Fault);
- hands-on PDK-based synthesis with a real 130nm process.

References

1. M. Bushnell and V. Agrawal, *Essentials of Electronic Testing for Digital, Memory and Mixed-Signal VLSI Circuits*, Springer, 2000.
2. IEEE Standard 1149.1-2013 — *Standard Test Access Port and Boundary-Scan Architecture*.
3. AUCOHL/Fault Documentation. <https://fault.readthedocs.io/>
4. Yosys Open SYnthesis Suite. <https://yosyshq.net/yosys/>
5. SkyWater SKY130 PDK. <https://github.com/google/skywater-pdk>
6. C. Wolf, *Yosys Manual*, 2004.

A Source Files and Commands

A.1 Source Files

<code>pattern_gen.sv</code>	<code>cut.sv</code>	<code>stuck_models.sv</code>
<code>MISR.sv</code>	<code>comparator.sv</code>	<code>controller.sv</code>
<code>scan_ff.sv</code>	<code>scan_reg.sv</code>	<code>d_ff.sv</code>
<code>top.sv</code>	<code>tb.sv</code>	

A.2 Generic Synthesis

```
yosys -p "read_verilog -sv cut.sv; synth; write_verilog -noattr -noexpr
cut_netlist.v"
```

A.3 SKY130 Synthesis

```
yosys -p "  
  read_verilog -sv top.sv;  
  synth -top bist_top;  
  dfflibmap -liberty sky130_fd_sc_hd_tt_025C_1v80.lib;  
  abc -liberty sky130_fd_sc_hd_tt_025C_1v80.lib;  
  write_verilog -noattr -noexpr bist_netlist_sky130.v;  
  stat -liberty sky130_fd_sc_hd_tt_025C_1v80.lib;  
"
```

A.4 Fault Simulation (baseline)

```
docker run -ti --rm -v "$PWD:$PWD" -w "$PWD" \  
ghcr.io/aucohl/fault:latest \  
fault -c simcells.v cut_netlist.v --clock clk \  
      --output-faultPoints faults.yml --output-covered covered.yml
```

A.5 Fault Simulation (LFSR replay)

```
docker run -ti --rm -v "$PWD:$PWD" -w "$PWD" \  
ghcr.io/aucohl/fault:latest \  
fault -c simcells.v cut_netlist.v --clock clk \  
      --externalTVSet lfsr_patterns.json \  
      --output-covered lfsr_covered.yml
```